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FOCUS ON RESEARCH METHODS

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Reliability in design science research

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Abstract

Reliability, which is concerned with whether something performs as it should, is well-accepted as an important part of any scientific research, with recognised methods for assessing it in the natural sciences. Reliability is a prerequisite for validity, which assesses knowledge production. Design science research is unique in that it produces an artefact and makes contributions to a design knowledge base and is tasked with the duality of design and science. Therefore, reliability should be reconsidered to address the assumptions underlying this type of information systems research.

This paper proposes a *Design Science Reliability Framework*. The framework identifies the part of design science research that should be evaluated (artefact, methods, measures and design knowledge). It also considers when reliability should be assessed as a project progresses by differentiating synchronic versus diachronic evaluation. The framework is used to derive a set of techniques for establishing reliability that can be applied during the evaluation process to both a project's knowledge production and its artefacts. Application of the framework should assist design science researchers when considering how to assess their results.

KEYWORDS

Design Science Reliability Framework, design science research, diachronic evaluation, reliability, synchronic evaluation, techniques for establishing reliability

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1 | INTRODUCTION

Design science research focuses on developing new ways to approach complex challenges in information systems by creating and evaluating artefacts that can solve real-world problems or improve existing solutions. It invents 'a new purposeful artefact to address a generalized type of problem' and evaluates it for solving problems of the same type (Venable & Baskerville, 2012, p. 142). This kind of research has a science component and a design component (Baskerville et al., 2015). Science is focused on analysis; design on synthesis (Simon, 1996). The science component requires a scientific approach to research. The design component recognises the important role of a designer who might be required to demonstrate creativity in solving a design problem, based on the production of an artefact. The two basic activities of design science research are build and evaluate. Building involves constructing an artefact for a specific purpose. Evaluation assesses how well an artefact performs (March & Smith, 1995, p.254). Evaluation is crucial and required to demonstrate the utility and quality of the artefacts and knowledge outcomes, as well as the rigour of the research process (Venable et al., 2016).

Design science research is pluralistic, contextual and dynamic, producing various types of knowledge as a study progresses (Akoka et al., 2022; Baskerville et al., 2015). A design is an abstraction of a problem that is in a designer's mind (Hooker, 2004), which may be difficult to extract and repeat. The objective of scientific research is to produce knowledge, so any academic study should be validated in order to make a contribution to science. For scientific knowledge, its quality is generally expressed using reliability and, subsequently, validity (Kirk & Miller, 1986). Different types of research require different types of validity criteria (Baskerville et al., 2023), yet the assessment of reliability and validity in design science is less common than the evaluation of efficacy or usefulness (Larsen et al., 2020; Prat et al., 2015).

Reliability is required to establish validity to ensure that research findings are accurate, consistent and trustworthy. Although reliability in design science research is assumed to be similar to reliability in other forms of science, it may not be as easy to establish as in the natural science, since the outcomes of design science research can take on different forms: a concrete, material artefact or generalised design knowledge. The outcomes can also depend upon the context. However, unless we can assess the reliability of design science research, it will continue to be difficult to convince researchers in other fields of the scientific validity of the results.

The question posed in this research is: '*What does reliability mean for design science research and how can it be achieved?*' Our objective is to investigate how to establish such reliability. To do so, we propose a *Design Science Reliability Framework* for identifying and assessing different forms of reliability in this type of research. The framework has three dimensions and provides a way to investigate different aspects of the design science research process. The first dimension is the 'what' of design science research (artefact, measures, method and design knowledge) and the second, the 'when' in the research process the research reliability is being assessed (synchronic versus diachronic). These two dimensions are used to derive the third, 'how' to establish reliability, presented as a set of eight possible techniques that could be helpful to researchers in formulating assessments of their studies. This, in turn, should help to justify the validity of the research. In this manner, this paper is intended to help focus design science researchers to avoid a reliability gap that occurs when validity measures lack reliability. To carry out the research, we follow an approach to the development of a conceptual framework (Jaakkola, 2020).

This paper proceeds as follows. Related research on design science and concepts related to reliability are reviewed in Section 2. Section 3 provides an overview of our methodology. Section 4 presents the *Design Science Reliability Framework*, with proposed techniques for establishing reliability detailed in Section 5. Section 6 discusses how this research extends prior work and Section 7 concludes the paper.

2 | RELATED RESEARCH

Table 1 describes the important concepts needed to understand the subsequent discussion and indicates how they relate to reliability as conceptualised and explored in this paper.

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TABLE 1 Description of terms.

Concept	Description and relationship to design science research reliability
Artefact	Human-creations designed during design research projects (Carvalho, 2012, p. 3)
Design knowledge	Knowledge produced as an outcome of design science research used to guide the building and evaluation of artefacts (Brendel & Muntermann, 2022, p. 1161); reliability of design knowledge should be established
Design science researcher	Researcher who conducts design science research project; one who would consider how to assess reliability and benefit from application of the Design Science Reliability Framework
Diachronic reliability	The 'extent to which the same observation made at different times yields the same information' (Kirk & Miller, 1986, p. 79); also known as external reliability and is one of the dimensions from which the 8 guidelines for establishing reliability in design science research are derived
Framework for Reliability in Design Science Research	Framework proposed in this paper; intended to support a design science researcher in assessing the reliability of their work. Comprised of: *What* is being assessed as reliable (artefact, method, (evaluation) measures, design knowledge); *When* (moment) reliability is assessed (synchronic or internal versus diachronic or external); *How* reliability is assessed (techniques for assessment)
Generalise	Work in a new setting other than the one in which something (e.g., an artefact) was created (Thomas et al., 2022). Establishing reliability contributes to confidence in the generalizability of the results from a design science research project; for example, from an individual problem (idiographic knowledge) to a class of problems (nomothetic knowledge)
Idiographic knowledge	Knowledge related to a specific problem; an individual design science research study
Knowledge base	Accumulation of design knowledge; design knowledge should be reliable as it accumulates into a knowledge base
Knowledge moment	Different points in time that occur throughout a project that have different assumptions and apply different design science research methods (Baskerville et al., 2015); results from each knowledge moment of a project should be reliable
Measure	Technique for assessment (evaluation); in design science research, the measures should be reliable
Method	Process or procedure for conducting research; in design science research a method needs to be reliable
Nomothetic knowledge	General concepts or theories for entire classes; helps a design science researcher to work from knowledge that has been established in order to develop new knowledge (Baskerville et al., 2015) that is reliable
Projectability	Application of design knowledge for very specific class of problems applied to a more general class; design knowledge must be reliable in order to be projectible
Reliability	• In science, generally refers to whether something performs as it should and in a repeated investigation (Gummesson, 1991) • In design science research, refers to whether design knowledge is useful in its original or future contexts. Challenged by dynamics of contexts, subjectivity in interpretation of outcomes and pluralism of design science
Reliability of measures	• Assessment of how reliable measures of observations or measurement scale are for problem-solving process; measurement procedure should yield the same result however and whenever carried out (Kirk & Miller, 1986, p. 19), which relates to dynamic and pluralistic nature of design science research • Distinct from the reliability of material artefacts and/or design knowledge and akin to that used in empirical studies

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TABLE 1 (Continued)

Concept	Description and relationship to design science research reliability
Types of reliability	Scientific reliability of knowledge outcomes; engineering reliability of artefact; pragmatic reliability (Germonprez et al., 2007) of design science research method and potential subsequent application of design theories produced
Validity	• Validity is the extent to which the credibility of findings and conclusions are established through evaluative procedures; systematic way to evaluate research claims (Cook et al., 1979; Guba & Lincoln, 1994) • Reliability is a prerequisite for validity in any scientific research including design science (Lincoln & Guba, 1985); encompassed by validity (Straub et al., 2004) • Research validity focuses on the quality of scientific research. It is important in design science because it provides procedural templates to collect and analyse evidence, and justify the arguments and conclusions of a study (Baskerville et al., 2015)

TABLE 2 Design science research characteristics.

Design science research	Relationship to this paper
Innovative and novel artefacts and design theories as research outcomes	• The Artefact should be reliable • Design theories are part of the design knowledge base • The Knowledge Base should be reliable
The construction and evaluation of artefacts constitute main research activities	• The Method used in the construction of the artefact should be reliable • The Measures used in the evaluation should be reliable since evaluation is considered a crucial part of design science research
From epistemological perspective, seeks utility over truth, rendering it pragmatic in nature	Focuses on how well an artefact works and its utility

2.1 | Design science research

Design science differs from other types of research because of the development of an artefact that must lead to additions to the (design) knowledge base. It is 'fundamentally different from both the theory-building and theory-testing approaches, which model themselves after the natural sciences and seek explanation based on observation' (Holmström et al., 2009, p. 67). Its characteristics have been described by Iivari (2015) and adapted by Brendel et al. (2021) as follows. (1) Design science research produces innovative and novel artefacts and design theories as its research outcomes. (2) The construction and evaluation of artefacts constitute the main research activities. (3) From an epistemological perspective, design science seeks utility over truth, rendering it pragmatic in nature. Table 2 summarises how these characteristics relate to design science reliability as presented in this paper.

As its name suggests, design science research is an amalgamation of design and science. It delivers symbiotic results in which design benefits from scientific rigour and science benefits from practical design grounding. Both the validity of new theoretical knowledge and the novelty of the design must be justified (Baskerville et al., 2017). Knowledge is created by analysing a given design problem, synthesising the possible solutions based on the analysis, and evaluating that solution.

To produce scientific knowledge, common expressions of the quality of such knowledge include validity and reliability (Carmines & Zeller, 1979; Kirk & Miller, 1986). Validity is concerned with whether an output from a scientific study is correct. (Gummesson, 2000). Reliability is concerned with the 'stability, dependability, stability, predictability

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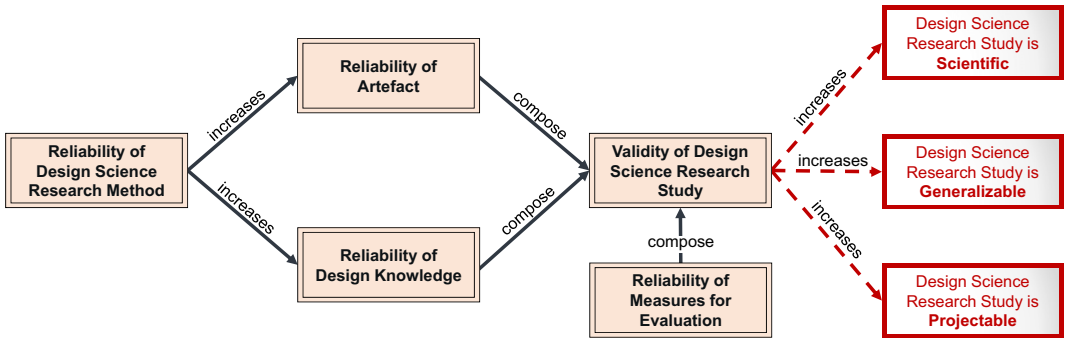


FIGURE 1 Role of reliability in design science research.

and accuracy of the results' (Kerlinger, 1973, p. 42). According to Boudreau et al. (2001, p. 2): 'reliability is a statement about measurement accuracy, i.e., the extent to which an instrument produces consistent or error-free result'. Since there is 'no validity without reliability', it is important to demonstrate reliability as a prerequisite for establishing validity (Lincoln & Guba, 1985, p. 42). Although research in information systems has studied validity in a variety of ways (Boudreau et al., 2001; Lukyanenko et al., 2015; Pillet et al., 2023; Schmitz & Storey, 2020), there has been much less effort given to studying the validity of design science research processes and outcomes (Larsen et al., 2020).

To achieve high scientific quality, generally, any measure used should be valid, reliable and replicable (Bateson & Martin, 2021). These three are distinct concepts (Garner, 2005), even though reliability has been described as to 'be about' repeatability and the 'extent to which measurements are reproduced when a study is repeated under the same conditions' (Bateson & Martin, 2021, p. 159). Reliability focuses on measurement accuracy (Boudreau et al., 2001; Rogers, 1995).

It is possible to replicate something, but it might not be reliable; for example, by using a scale that is calibrated 3 kg too high or too low.[1] Repeated applications of the scale will consistently provide relative, but not accurate, results. Another example is a tuning artefact used to tune an orchestra that is not at standard A440 pitch. This would be extremely problematic for musicians with perfect pitch or geographically dispersed orchestra members playing simultaneously, as in a virtual performance. These artefacts are replicable but not reliable. In contrast, an explosive device artefact could be applied to implode a building. It is reliable for one instance, but not replicable. In design science research, something might be both replicable and reliable. Brendel and Muntermann (2022), for example, focus on the *replication of design theories* (one type of design knowledge). Rosenkranz et al. (2017) report on repeated applications (multiple replications) of their artefact. In general, replicability is expensive for design science research, so one way to view our work is, essentially, as a way to reduce such an expense.

Figure 1 summarises the role of reliability in design science research.

Design science research fundamentally creates an artefact using a method. The selection and execution of the method impact the reliability of both the artefact and resulting design knowledge (e.g., principles and theories). Thus, we start by considering the reliability of the method and how it increases the reliability of both the artefact and design knowledge. Since reliability is required for validity, the reliability of the artefact and design knowledge, as required outputs, impact the validity of the study. Evaluation of the artefact is a further requirement, so the measures used in the evaluation must also be reliable.

[1]De Schryver et al. (2016). Unreliable yet still replicable: A comment on LeBel and Paunonen (2011). *Frontiers in psychology*, *6*, 173 501. 'There is no direct, fixed or one-to-one relation between reliability and power or replicability' (p. 1).

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Thus, to validate a design science research study, the reliability of the artefact and design knowledge are required, as well as the method used to create the artefact and the measures used in the critical evaluation. The validity of a design science research project increases confidence in the design science study being scientific and having the desired generalizability. Only then, can the results can be projected to other projects, consistent with Simon (1969)'s argument that a design is useful only when it can solve related problems. In this way, reliability affects the generalisation, projectability and scientific quality.

2.2 | Role of reliability

Reliability in design science research is different from behavioural research and can also be established by other means (e.g., mathematical proof or logical deduction, or, perhaps, non-tangible means such as aesthetics (Baskerville et al., 2018)). The design knowledge (abstract solutions) created by this research plays a more important role in engineering practise than theories in social science practise and tends to be more complex. Engineering involves formulating a problem in such a way that it can be solved through design. Science focuses on formulating a question that can be solved through investigation. The design process of engineering is somewhat similar to that of the scientific method (Dieter & Schmidt, 2009). Both begin with existing knowledge, and gradually become more specific as they search for knowledge (in pure or basic science) or for a solution (in applied science). The main difference is that the engineering process focuses on design, creativity and innovation, whereas the scientific process emphasises discovery.[2]

There is a difference between the reliability of design science research and reliability in empirical research, since design science can be anchored in the artefact; specifically, in the instance solution (Lee et al., 2011). The artefact itself can be reliable. Reliability and validity are features of a study, implying an empirical context and empirical data. For example, a mathematical proof (potentially based on the law of physics) may be at least as good at establishing reliable relationships between different constructs. Therefore, establishing the technical reliability of an artefact (e.g., that software is safe for landing an aircraft) could apply a mixture of (empirical) tests and mathematical proofs.

The engineering and computer science disciplines have established practises to evaluate reliability, which include methods for artefact creation and measures for evaluation. An automobile producer, for example, primarily establishes the reliability of the parts delivered by its suppliers by auditing adherence to quality norms (e.g., ISO 9000) that check whether a structured process exists and is adequately documented. Automobile producers also have test strategies for checking reliability. They have different sampling strategies in that they typically focus on edge cases, rather than repeatedly evaluating standard cases. This is not so in standard empirical studies or design science evaluations.[3] There are also empirical evaluations of the artefact's utility. The strive for utility is implemented by defining a meaningful scope and then measuring it.

In other engineering efforts, reliability is a feature of an artefact that advances reliance; that is, it advances an observable human behaviour. Validity and replicability are features of a study such as that used in the evaluation. Research in information systems, generally, focuses on more than just technical artefacts; it focuses on socio-technical systems (which is beyond the scope of this research). Analogously, design science focuses on more than the material artefact. It also has requirements for the production of design knowledge and an evaluation component.

There is a great deal of interest on assessing reliance on automation and artificial intelligence (AI), including generative AI (GenAI). There are also efforts to understand human trust in AI (Lukyanenko et al., 2022; Oyekunle et al., 2024). Reliance is an observable behaviour, in contrast to trust, which is an attitude that cannot be directly observed. It may be argued that reliability is a feature of an artefact that advances reliance; validity, reliability and replicability are features of a study.

[2]See https://en.wikipedia.org/wiki/Engineering_design_process, in addition to Dieter and Schmidt (2009) and Ralph and Wand (2009) for a detailed examination of the general concept of design.

[3]Other typical evaluation strategies separate the test of sub-components from integration tests, and so on.

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An outcome of design science research is a contribution to the design knowledge base. Lee et al. (2011) stress the importance of theory in design science research, emphasising that *design theories* (as part of the knowledge base) have specific components (meta-requirements, meta-design, design method, testable product hypotheses, testable process hypotheses, etc.). Design theories 'encompass a design process for applying a meta-design for the purposes of instantiating a designed artifact' (Lee et al., 2011, p. 2). Lee et al. (2011) argue that the study of the instance problem is frequently similar to a classical (qualitative) empirical study (e.g., ethnography or experiments), even though the outcome variables are utility-based.

The abstract problem is grounded in some theory, and thus, reliability could follow the standards of behavioural research. The abstract solution (most importantly, the design principles) is typically described as an architecture or an algorithm (similar to engineering). This is part of the design knowledge produced. According to Walls et al. (1992) this could be technology rules, or other forms. Much research has focused on the abstraction of design principles or design features or testable propositions.

2.3 | Problem solving research

Design science research is a problem-solving paradigm that focuses on providing prescriptive solutions that serve human purposes through the production of artefacts (March & Smith, 1995), and often conducted in an iterative manner (Baskerville et al., 2023). The difference between the reliability of design research and reliability in empirical research can be anchored in the artefact, specifically, in the instance solution (Lee et al., 2011). The artefact can itself be reliable. Validity and replicability are features of a study that have an empirical context and empirical data.

Contributions come from both the artefact produced and the abstract, design knowledge, found in (design) knowledge base of principles, theories, and so on (Baskerville et al., 2015; Woo et al., 2014). As explained by Sonnenberg and Vom Brocke (2012, p. 384), the evaluation of the artefacts should be 'approached differentially compared to the study and evaluation of phenomena in the natural or human sciences'. This is because the scope and interest of design science research is not to explain how the world works through observations, theories, and so on, but rather 'to shape the world by means of artefacts'. They further note that evaluations need to account for the emergent nature of the research and decisions made by the designers, which has implications for the 'validity of knowledge created while the artifact emerges'.

2.4 | Artefact reliability

Design science research produces both an artefact and design knowledge. The reliability of an artefact requires it to function in a predictable way; for example, a machine learning model should behave in the same way given different data. Artefact reliability can also be dependent upon its goal or situation, which might be dynamic. This is because a design science study is pluralist, experiencing different knowledge moments as it progresses (Baskerville & Myers, 2015). While law-like, design knowledge falls short of natural laws. Imperatives, rather than causals, characterise design knowledge: Do *a* and *b* happen (not simply *a* causes *b*)? In practise, both *a* and *b* can be complex events. In a future context, some variation of *a* may be sufficient for *b* to happen. It is the present and future usefulness of design knowledge imperatives that establishes reliability (and validity). Consequently, the goals of design knowledge are generative and inventive, even though they must adhere to variances in requirements and constraints (Baskerville et al., 2015).

In information systems research, there is a well-recognised need for both rigour and utility. Validity, broadly, deals with the extent to which the credibility of findings and conclusions are established through evaluative procedures (Trochim, 2006). In design science, validity refers to how well an artefact performs (March & Smith, 1995) and is a measure of whether the results from the research are correct. Validity can be crippled by low reliability because

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results that are unreliable cannot subsequently justify validity. Reliability must be established for both realised and realisable generalisations, a quality known as *projectability*, which relates to how the new research context and goals align with those of the projects from the knowledge base (Baskerville & Pries-Heje, 2014; Baskerville & Pries-Heje, 2019).

Evaluation is important. Design science research results are typically anchored to the evaluation of an artefact.[4] If an artefact is unreliable, we cannot estimate the reliability of any evaluation measures, and, subsequently, the validity of any knowledge produced, including design principles or theories. Similarly, the method that produces the artefact must be reliable (Peffers et al., 2007). Reliable methods ensure that the design science research is indeed scientific.

2.5 | Additional perspectives and assumptions

Traditional conceptualizations of reliability often rest upon assumptions of repeatability, replicability and consistency (Lincoln & Guba, 1985). However, in design science, we must consider a dynamic view of the world, where situations change and the outcome can be subjective, depending upon the purpose or types of design knowledge and artefacts.

The literature in science and engineering provides several forms of reliability estimates that can be applied to design science artefacts. These include *measures*, *methods* and *design knowledge*. Since design science research depends on both the creation of an artefact and the subsequent contributions to design knowledge, it is important to consider each of these when assessing the reliability of a design science research artefact. The reliability of design knowledge should depend on the accuracy and completeness of the conveyed knowledge. Accuracy means that design knowledge correctly transfers the design principles and/or theory. Completeness means that the design knowledge extends to cover the context(s) in which the design is known to operate. Completeness is necessary because a designer seeking to apply this knowledge in a different context needs to know enough about the original context(s) to adopt or adapt the knowledge as appropriate for a new or different context.

Design theory is divided into two types of knowledge: product (meta design) and process (design process) (Walls et al., 1992). The former is the design principles/theory and the latter, the adaptation process. Design knowledge accuracy without completeness (or vice versa) cannot be reliably reused. Research produces knowledge. Reliability characterises knowledge, and reliability characterises the artefacts produced by design science research.

Especially in qualitative methods, reliability is challenging and entails a repeated investigation that is expected to obtain the same result (Gummesson, 1991). In research approaches such as case studies, ethnography, and grounded theory, both the subjects and the situation or context within which the subjects appear naturally change over time. Establishing reliability is even more difficult with interventionist methods, such as action and design science research, because these methods purposefully introduce social changes or new artefacts into the research settings. Then, the concept of *dependability* provides an alternative to reliability (Guba, 1981) because it values the trustworthiness of knowledge instead of its truth value. Most social or design interventions do not have clear quantitative evidence found in other kinds of research. The outcome measures are often loose notions of artefact properties; for example, whether the task was completed with a valuable impact (Venable et al., 2016) or the dependability of an artefact in its capacity to perform failure-free in different use conditions (Piccoli et al., 2020).

[4]'We evaluate artefacts to determine if we have made any progress. The basic question is, how well does it work. … Evaluation requires the development of metrics and the measurement of artefacts according to those metrics. Metrics define what we are trying to accomplish. They are used to assess the performance of an artefact. Lack of metrics and failure to measure artefact performance according to established criteria result in an inability to effectively judge research efforts' (March & Smith, 1995, p. 258).

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2.6 | Prior reporting

To investigate reliability and related concepts, we identified approximately 140 design science research papers from the AIS Basket of Eight journals, published between 2004 and 2020. The Basket of Eight is well-recognised as providing important outlets for disseminating research in design science. It is commonly used as a representative sample of quality publications when analysing the information systems (IS) field (Gregor et al., 2020; Johannesson & Perjons, 2014). Of the papers obtained, approximately 60 mentioned reliability, although with inconsistent use and discussion. Table A1 (Appendix) summarises the papers.

In some of the papers, the terms are simply mentioned for various reasons, but not directly discussed within the given context. For example, Arazy et al. (2010) propose a methodology for theory-driven design that they applied to social recommender systems to improve prediction accuracy. They investigated their design based on a behavioural study regarding people's likelihood of advice-taking. In their publication, the researchers reported on the reliability of the experimental constructs, but not of the system design.[5]

Abbasi et al. (2012) refer to stacked generalisation to improve reliability. Bygstad (2017) distinguish lightweight information technology (smart phones, tablets, etc.) from heavyweight technology, reporting that the latter requires reliability. Coenen et al. (2018) argue for the need to evaluate competencies stating that the common rubric approach is harmful for both validity and reliability. Ebel et al. (2016) report inter-rater reliability for business model case studies. Fahmideh et al. (2019) included interviews in a broad effort to confirm validity of secondary documents for case studies.

Yang et al. (2012) propose a win–win negotiation agent with two strategies (simultaneous-equivalent offers and delayed acceptance) and address reliability with respect to their experimental design. Liu et al. (2020) propose a text analytic framework to extract features from online forums and then to classify them based on the usefulness of their solution. Their evaluation efforts are based on two online, large-scale communities (Oracle and Apple). They discuss the training of domain experts on the manual annotation of the forms for reliability, and use standard information retrieval metrics for assessment of their framework. They also discuss how to improve the validity of their framework. Piccoli et al. (2020) address both reliability and validity for the development of their digital skills mastery artefact. Other efforts simply identify the need for validity or reliability, but do not address it directly. Many of the mentions of reliability were in passing and not related specifically to the artefact or design knowledge or methods of the research or evaluation. Overall, there is a lack of consistent effort to include reliability in evaluation discussions.

2.7 | Related research efforts

Other research efforts are somewhat related.[6] De Leoz and Petter (2018), for example, consider the social impacts of artefacts in design science research. They conclude that the evaluation of an artefact is usually performed on its technical ability and utility. They argue that, in artefact evaluation, the social impacts within the environment should be considered, with an ex post evaluation to identify the impact of the artefact within the social structure (guideline 4, p. 161).

Iivari et al. (2021) focus on design principles, suggesting that, for papers whose output is primarily design principles, they should be able to evaluate their reusability and propose a framework for doing so as a contribution to the external design knowledge base. Nunamaker et al. (2015) propose that the artefacts actually developed should be implemented in the field, which might involve trading off rigour for relevance. They refer to this as a credibility assessment. Pentland and Feldman (2008, p. 235) discuss the 'role of artifacts in shaping organizational routines' and suggest how to design organisational systems as generative systems. Prat et al. (2015) investigate the *what* and the *how* of IS artefact evaluation. The *what* is the objects that are used in evaluation and the how is the method to evaluate them. They create a taxonomy of evaluations as applied to design science research from the Basket of Eight.

[5]Additional personal correspondence with the lead author.

[6]The authors thank the anonymous reviewer for suggesting this research.

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2.8 | Synchronic versus diachronic

An important consideration related to reliability assessment is the time at which the assessment occurs, which we classify as synchronic or diachronic.

2.8.1 | Synchronic reliability

The assessment of synchronic reliability, or internal reliability is associated with single cases. Synchronic artefact reliability is important because many design settings are idiographic: concrete, individual or unique (Baskerville et al., 2015). Design settings may also have social, cultural and behavioural aspects. Then, idiographic accounts can be used: narratives and understanding instead of observations and objectivity as the empirical means to assess artefact reliability. It is an attempt to gain an understanding of a situation by 'obtaining first-hand knowledge of the subject under investigation' (Burrell & Morgan, 1979, p. 6).

Synchronic reliability helps researchers reconcile differences in multiple, simultaneous observations (Kirk & Miller, 1986). Synchronic reliability tests provide an opportunity to evaluate the reliability of related aspects at the same time, such as the performance of an artefact in two different, but simultaneous, settings. It can also be applied to different issues of a study: individualised characteristics; causal relationships of variables within a single case; idiographic predictions based on patterns of data about a single case; specific subjective analysis of a certain case; and identification of the central themes in a certain case (Runyan, 1983).

Techniques for determining the synchronic reliability include comparing data that is acquired through different forms such as split-half testing or inter-rater correlations (Kirk & Miller, 1986). For example, two or more observers could observe the same phenomenon during the same time frame. Alternately, a researcher could use a number of different chains of evidence from the same case study. Synchronic reliability is similar to triangulation in the sense that it uses simultaneous measures of the same event.

2.8.2 | Diachronic reliability

Diachronic reliability is the 'extent to which the same observation made at different times yields the same information' (Kirk & Miller, 1986, p. 79) and is also known as external reliability. Diachronic reliability assessments are generally associated with multiple cases. This association is not absolute. Diachronic reliability assessments may be associated with either the stability, or evolution, of a single case.

In assessing diachronic reliability, a specific type of observation is expected to yield the same measurement when exposed to a particular phenomenon at different points in time. It focuses on the stability of an observation over time, which is the 'similarity of measurements or findings taken at different times' (Baskerville, 1996, p. 18). Techniques for evaluating diachronic reliability include conducting repeated experiments or examining the changes over a period of time. Barley (1990), for example, illustrated diachronic analysis by contrasting the use of a single technology in a hospital setting during an earlier and a later time period.

3 | METHODOLOGY

Figure 2 provides an overview of our approach to the development of a conceptual framework, which is intended to serve as a useful way to understand reliability in design science, and how design science researchers can systematically apply it. The framework is conceptual in that it does not provide testable results, but aids in understand a field (boundaries, finding, challenges) (Matook & Brown, 2017; Shapira, 2011). It could serve as a template (Peffers et al., 2007) for evaluating the results obtained and a classification (Matook & Brown, 2017).

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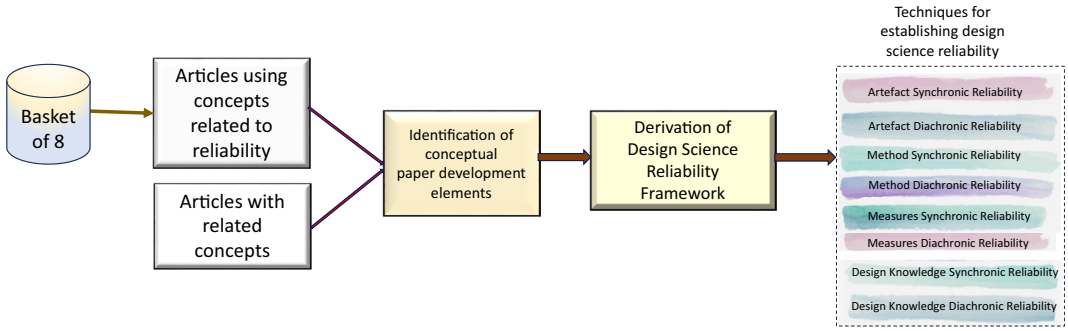


FIGURE 2 Overview of research approach.

3.1 | Conceptual paper approach

We followed the suggested approach to writing a conceptual paper (Jaakkola, 2020) to show how we derived the Science Reliability Framework and classify techniques for establishing reliability. The framework's development was based on logic and research on design science as well as relevant research in other areas, as shown in Figure 2.

We, thus, propose a framework, developed as a conceptual piece of research, following the elements identified by (Jaakkola, 2020). The research domain is design science research. The concept under investigation is reliability in this domain, noting that reliability is a well-recognised, important concept in scientific research, but much less investigated in design science than desirable in the evaluation process. Conceptual papers are often based on multiple concepts (Figure 1), the literature (AIS Senior Scholars' Basket of Eight and related papers) and theories (prior contributions to design science research). Table 3 summarises the application of the main elements to our work.

As in all conceptual papers, the literature review was important to understand the state of reliability assessment in design science research and to consider empirical assessment. The framework resembles a typology (Delbridge & Fiss, 2013; Doty & Glick, 1994) because it distinguishes, dimensionalizes or categorises the knowledge of the phenomenon. Here, we provide the dimensions (what and when) of the various types of the phenomena of interest in the domain of design science to create a set of techniques (how) to address reliability.

3.2 | Review of existing literature

For the conceptual development of the framework, we reviewed existing literature and how it was used (Avdiji & Winter, 2019). Identifying a design science research paper is not always straightforward. To ensure we captured design science papers, we started by extracted papers from the intersection of those from the AIS Senior Scholars' Basket of Eight journals as identified by Prat et al. (2015) and those identified by Deng et al. (2017) in their systematic literature review to examine theoretical and empirical design science research. We then identified design science research articles that were published between 2004 and 2020 in the AIS Basket of Eight journals, which are recognised as important outlets for disseminating information systems research. Since the *International Conference on Information Systems* (*ICIS*) is the premier conference in the information systems field, we searched for all papers on design science research that were published in this conference, the result of which identified 99 additional articles. We also consulted articles with relevant, related concepts.[7] The articles were searched for mention of the terms

[7]We appreciate the efforts of a reviewer to suggest these articles.

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TABLE 3 Conceptual paper elements (based on Jaakkola (2020)).

Conceptual paper element	Application to this paper
Choice of theories or concepts used to generate insights	Focal phenomenon observable but not addressed in literature: reliability in design science research Focal theory or other theories to bridge observable gaps: positioning with respect to validity in scientific studies
Choice of theories and concepts analysed	Design science methodological research
Perspective; level(s) of analysis/aggregation	General techniques for design science researchers
Key concepts to be analysed/explained or used to analyse/explain	Concepts of reliability related to design science research artefacts and knowledge base; diachronic versus synchronic reliability
Translation of target phenomenon in conceptual language; definitions of key concepts	Target phenomenon is reliability; supporting terms described in Table 1
Approach to integrating concepts; argumentation	The 'what' (measures, method, design knowledge) and 'when' (synchronic, diachronic) dimensions influence the 'how' to lead to derivation of 8 techniques to assess reliability

reliability, reliability, validity and repeatability, or their derivatives. This was followed by manual investigation of their usage within these articles.

4 | DESIGN SCIENCE RELIABILITY FRAMEWORK

The *Design Science Reliability Framework* is presented in Figure 3. The framework development involved: the literature on reliability; known, unique characteristics of design science research; and prior work. The framework captures the knowledge dimensions (what is reliable) and the knowledge moments (when reliability is assessed). The 'what' (artefact, method, measures and design knowledge) and the 'when' (synchronic versus diachronic), lead to the identification of 8 ways (the 'how') in which design science studies can demonstrate reliability that are important for justifying validity.

The framework focuses specifically on the assumption that design science research attempts to address a generalised type of problem. However, this is only one strategy for conducting this type of research. Others focus on addressing specific problems and then starting to generalise retrospectively after several cases have been documented (Iivari, 2015; Rohde et al., 2017).

Reliability in design science is more multifaceted than in other research paradigms. It involves *what* is being regarded as reliable, *when* it is regarded as reliable and *how* it is regarded as reliable (Butler & Gray, 2006). W*hat* is reliable: the artefact itself, often considered in terms of its function and utility, but here, also, the method to develop it, measures to evaluate it, and contribution to the knowledge base produced from it. *When* these are reliable: at what knowledge moment in the research does this artefact reliability assessment take place? *How* it is reliable: the techniques used in its assessment in the evaluation process.

Reliability can be conceptualised as a criterion closely connected to viewing design knowledge as nomothetic, but imperative, knowledge. Nomothetic knowledge refers to general theories or concepts that cover the entire set of classes of a given case, in contrast to idiographic knowledge obtained from situated artefacts (Baskerville et al., 2015).[8] Design knowledge, as nomothetic knowledge, is reliable if it can be used in the same context again.

[8]'Design-science researchers derive nomothetic knowledge through processes which involve abstract thinking that considers the kind of problem at hand or the kind of solution that might be effective. Nomothetic processes also help the researcher work from previously established knowledge to develop new knowledge' (Baskerville et al., 2015, p. 548).

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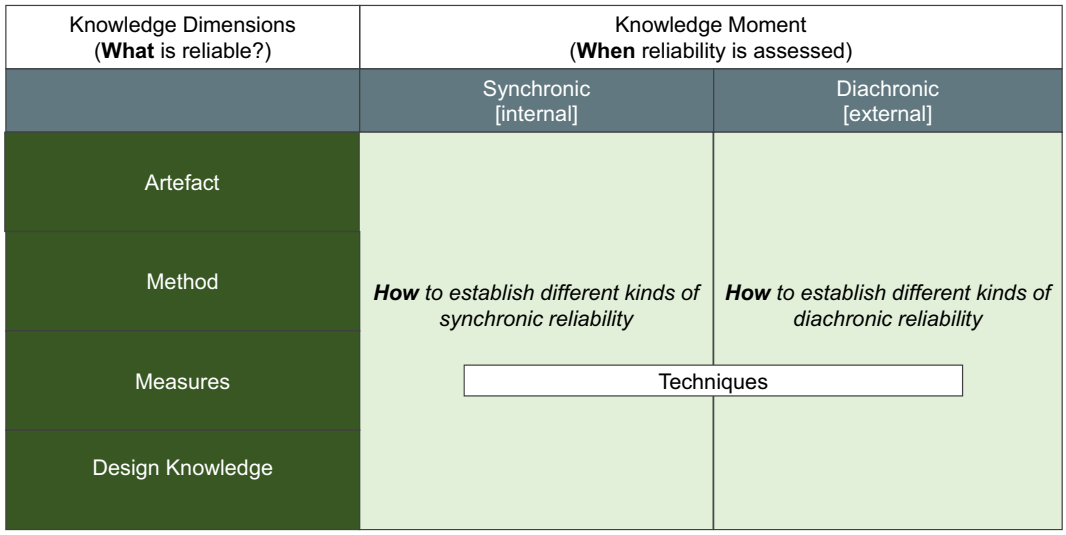


FIGURE 3 Design Science Reliability Framework.

However, contexts are frequently dynamic, so the possibility to apply it to other contexts, and to be useful at another time, becomes critical to establish design knowledge as reliable. It is this conceptualization of design science research that led to the derivation of our framework.

Furthermore, because design science research attempts to address wicked, real-world problems, there can be some subjectivity in any design science outcome. Design science research has an iterative process, performed by researchers who have their own individual style and capabilities. The outcome of a project depends on the context, purpose and types of design knowledge, and the artefacts produced. Therefore, making the situation, or context and design choices as explicit as possible, should make it feasible to achieve a shared understanding among researchers regarding their perceptions of the outcomes when idiographic knowledge is produced. This, in essence, can align the mental models of researchers and make the outcomes comparable on an abstract level, while acknowledging both subjectivity and interpretations and still striving to find general outcomes.

The distinction between nomothetic and idiographic reflects the need for an artefact, as an important outcome of a design science research study, to behave in a predictable way (nomothetic knowledge) versus for a given context (idiographic knowledge). The latter is due to the acknowledged pluralism of design science research that impacts the evaluation process so that different criteria are needed to evaluate different parts of a study.[9]

4.1 | 'What' dimension

Researchers must, generally, be concerned with the reliability of the output: the artefact created and the design knowledge extracted. Also important are the reliability of the method used to create the artefact and the measures used to evaluate it. The *what* dimension thus identifies the four assessable scientific aspects of design science research: (1) the reliability of the *artefact* created; (2) the reliability of the *method* used to create the artefact; (3) the reliability of the *measures* applied to evaluate the artefacts; and (4) the reliability of the *design knowledge* embedded within and extracted from, the artefact.

[9]Scientific pluralism refers to different theories, approaches, values and perspectives in scientific research. Kellert et al. (2006). *Scientific Pluralism* (Vol. 19). U of Minnesota Press.

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4.1.1 | Artefact reliability

Artefact reliability is more than the dependability of an artefact's functionality. It is also dependent upon the design process, the manner in which observations are made, and the application of the design scientist's knowledge and creative thinking. Many claims in design science research findings assume reliability of the material artefact in delivering its function and utility. Does the artefact do what it is supposed to do within the built environment in a dependable way? Is it robust in working as intended? The reliability of an artefact ideally aligns with the dependability or robustness in the way the artefact behaves in the 'build-and-evaluate' cycle. It is not just 'how well the artefact performs' (March & Smith, 1995, p. 254). It is whether it performs reliably in ideal circumstances, or even in less-ideal circumstances. The concept of reliability in design science artefacts, then, is similar to the concept of reliability in engineering. For engineers, the end-goal is functional performance. This view of artefact reliability is ideal for material artefacts. This view also inhabits conceptualization of artefact reliability for other kinds of design science artefacts: constructs, models, methods and instantiations, design patterns, design propositions and so forth.

4.1.2 | Method reliability

Method reliability is concerned with the need to understand and evaluate a method (Thoring et al., 2023). It ensures that the method for conducting the research leads to valid and consistent results. There are widely-used design science research methods, such as that proposed by Peffers et al. (2008).

4.1.3 | Measures reliability

The reliability of the evaluation method provides assurance that the artefact evaluation is conducted in ways the scientific community will recognise and accept. Reliability of measures refers to any measurement scale or measures of observations. For data collection, design science evaluation is similar to other forms of science, where empirical measures and observations are subject to assessments of reliability. The centrality of measure or observation reliability is perhaps most characteristic of a quantitative context, where instrument reliability is critical to accurately reflect the phenomena under observation (Carmines & Zeller, 1979).

4.1.4 | Design knowledge reliability

Design knowledge reliability is crucial in design science research because design knowledge focuses on knowledge that is intrinsic to the design and development of an artefact. Design knowledge outcomes can come from both the creation of new kinds of artefacts and the process of designing them.

The researcher's knowledge provides the foundation for an artefact, as well as theoretical knowledge necessarily embedded within it. Therefore, the reliability of the researcher's knowledge is critical to the functioning and utility of the artefact. The researcher brings his/her knowledge, experience and creative thinking into the design process. Design knowledge usually includes specialised knowledge of a design domain (management, finance, manufacturing, government, etc.). Regardless, there is an underlying axiom that acknowledges forms of knowledge special to the awareness and ability of a designer (Cross, 1982, p. 224). In engineering design, knowledge concepts are generated from inventions, theoretical engineering research and experimentation, whereas practical considerations arise from design practise, production and trial (Vincenti, 1990).

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4.2 | 'When' dimension

A design science research project has *knowledge moments*, through which it might pass on its way to completion, making the research process a dynamic one (Baskerville et al., 2015). Akoka et al. (2022) propose *knowledge paths*, to represent the knowledge contribution of a single project as a succession of knowledge moments, characterised by a project's goal and scope. The *when* dimension (knowledge moment) is shaped by the iterative nature of design science research, reflecting its pluralistic nature. Reasons for iterations include: the complexity of the problem (Peffers et al., 2007); the aspect of search that yields the interfacing artefact between the inner and outer environments (Simon, 1996; Thomas et al., 2022), and the process of evaluation and circumscription (Kuechler & Vaishnavi, 2012).

Justification for *when* to evaluate artefact reliability can involve investigations that occur at a specific point in time (synchronic) or over a period of time (diachronic) (Swinburne, 2001). The concept of time in design science research can be more flexible, referring to the extent of the similarity of observations within the same general time period.

4.3 | 'How' reliability established

The two dimensions of *what* is being assessed, and *when* the assessment is being conducted, help to understand *how* a reliability assessment can be achieved. The *what is* concerned with what is being characterised as reliable: artefact, method, measures or design knowledge. The *when* is the temporal relationship between reliability and time. The *how* dimension varies with the *what* and *when* and used to explore and evaluate artefact reliability on both dimensions of the framework.

5 | EMERGENCE OF TECHNIQUES FOR ESTABLISHING ARTEFACT RELIABILITY

From the dimensions of *what* (three components) and *when* (two components), the *how* emerges as the eight (4×2) possible reliability techniques shown in Table 4. These techniques should assist researchers and reviewers in assessing the type of reliability appropriate for different aspects of a project, thereby contributing to its scientific aspect. The specific examples might serve as potential reference points.

5.1 | Artefact: Synchronic reliability

If an artefact is claimed to be reliable for a single context, then, reliability can be shown by assessing the utility of the artefact in its original environment. Silic and Lowry (2020), for example, conduct a design science research study for gamification. They were concerned with design evaluation for proof-of-value, for which they used a controlled field experiment. The proof-of-value is intended to show that the IT artefact actually works in reality. They were able to apply selected design principles identified in a related project (Liu et al., 2017).

5.2 | Artefact: Diachronic reliability

To claim an artefact as reliable for multiple contexts, it is important to show reliability with repeated assessments of the artefact's performance and utility independent of the original environment in which the artefact was developed. Rosenkranz et al. (2017), in an extensive study of data warehouse development, summarise their evaluation as

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TABLE 4 Techniques for establishing different types of reliability.

What	When: Synchronic	When: Diachronic
Artefact	• If claimed reliable for a single context • Show reliability by assessing the utility of the artefact in an original environment • *Example*: (Silic & Lowry, 2020)	• If claimed reliable for multiple contexts • Show reliability with repeated assessments of the artefact's performance and utility independent of the original environment • *Example*: (Rosenkranz et al., 2017)
Method	• Show reliability with method that assesses the utility of the artefact in a multi-level environment or an environment that is contra-ideal • *Example*: Parsons and Wand (2008); Gottschalk et al. (2022)	• Show reliability with method using repeated assessments of artefact's performance and utility in environments other than the original design setting • *Example*: Anderson et al. (2023); Lacity et al. (2021)
Measures	• Always multiple kinds of measures • Triangulate with multiple, different measures at the same time • Show artefact reliability with singular, but intensive, application of the artefact that produces significant results from multiple observers, multi-level observations or multiple sources of evidence • *Example*: Abbasi and Chen (2008); Albert et al. (2004), Scholta et al. (2020)	• Singular or multiple kinds of measures • Use the same (set of) measure(s) at different times • Show artefact reliability with multiple instances of measurements or observations at different times and/or different non-synchronous settings • Show multiple applications of the artefact that produce significantly desirable results, ideally by different researchers, for different problems at different times in different settings • *Example*: Dincelli and Chengalur-Smith (2020); Piccoli et al. (2020); Akello et al. (2023)
Design knowledge	• Show practical significance of the artefact through the consistency of high utility discoveries as a design process progresses, especially in cases of iterative design such as prototyping • *Example*: Kohler et al. (2011)	• Show practical significance of artefact by the iterative and evolving usefulness of the artefact as a model (design knowledge) across other settings, other design problems, and/or at other times • *Example*: (Pries-Heje & Baskerville, 2008)

follows (p. 98). Their assessment of validity and utility showed that the artefacts supported the development of both process and content common ground in the design of data integration requirements. Their expectations regarding observable behaviour in real-world data warehouse development projects were confirmed by case studies, which showed that their artefacts—template, procedure model and data requirements tool—led to early detection of communication defects and data integration errors. The case studies and surveys showed that the artefacts enhanced project performance and increased quality. They recognise reliability in the need to document the procedures used in their cases for others to follow.

5.3 | Method: Synchronic reliability

Synchronic reliability of a method means that a method's application at a specific time, and within a specific context, yields significant results. It is a form of a one-time reliability, established through significant scientific outcomes from a singular study in which the intensiveness of the study propels results so forcefully that reliability concerns about errors or distortions from biases are virtually eliminated. Yin (1994) justifies the belief in idiographic results based on multiple sources of evidence, such as a case study database with a structured survey, open-ended interviews, protocol-driven observations and organisational documents. Such preponderance of evidence justifies the reliability of the results through

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the intensiveness and forcefulness of a study's evidence. Parsons and Wand (2008), for example, consulted a panel of experts in a protocol-driven analysis to evaluate their proposed set of classification rules, which were derived from cognitive notions of inference and economy. Gottschalk et al. (2022) propose a crowd-validation process when designing a software artefact, although they used three design cycles to obtain the design knowledge and implementation.

5.4 | Method: Diachronic reliability

Diachronic reliability exercises the method used by different researchers in a different place at a different time. Alternately, it could be the use of a specific method over a long period of time through a prolonged engagement as in action design research (Sein et al., 2011). It assumes that a method will remain stable under different research contexts. Since the context may differ, it is likely that repeating the research at a different time, or within a different context, may yield differing results. Therefore, in assessing the reliability of a method across different contexts, the results from using the method might differ somewhat. It then becomes important to assess whether the method worked as intended, did not work in a particular situation, or requires modification.

This kind of reliability is established by demonstrating significant scientific outcomes, when the same method is used in multiple studies, different research cases or settings, different researchers and at different times. A method can be considered reliable, even when it must be modified to fit another setting. Such reliability depends on showing a history of significant usage, although it can take considerable time to establish this form of reliability.

The design science research methods of Peffers et al. (2007) and Vaishnavi and Kuechler (2015) form the basis of many published studies. A prior study (Baskerville et al., 2017) reviewed articles that specifically referenced the method proposed by Peffers et al. (2007). As a scientific method, it is reliable because it guides design science research that delivers scientific contributions that are novel and substantial. A variety of applications of the method suggest that the method has a certain robustness and flexibility that permit its adaptation by significantly different researchers in different settings and times. Publication of these efforts in leading information systems journals further demonstrates the reliability of the method as robust and adaptable.

Lacity et al. (2021) formalise action principles for new technologies, while complementing design science and action research. Their work is based on methods related to sourcing, business processes, automation, blockchain and possibly others. In their development of action principles, they considered 22 cases across multiple contexts.

Anderson et al. (2023) take a design science research approach to address the problem of patients' self-management of consent of online data sharing. The researchers evaluated the artefact, DApp, through simulations using validated measures of latency and throughput, demonstrating practical or pragmatic utility and reliability, as well as engineering or material artefact reliability. If the researchers had developed a prototype without articulating their efforts to establish the reliability of the outcomes, then the research might have demonstrated a novel idea, but might not have been able to effectively establish rigour and relevance. The researchers describe their method as using two iterations based on Peffers et al. (2007) and distinguish how the artefact was different from that produced in an idiographic project. From their method, the researchers established design principles and design features. This abstraction approach resulted in a design that could be projected beyond the original problem of patient consent to any kind of personal data management. Using a well-established design science method provided rigour and enabled projectability of the artefact. The project progressed from a specific problem to general design principles; and from a specific artefact to a generalizable artefact design adaptable to other problems in the same class.

5.5 | Measures: Synchronic reliability

Synchronic reliability of an artefact means evaluating an artefact at a certain point in time instead of evaluating the performance of the artefact over time. It can be established by different ways of measuring or observing at the same

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time or in the same setting. The timing helps to establish artefact reliability. Using first-hand measures (e.g., observing what people do, rather than asking them what they did) diminishes the effects of self-reflection on reliability measures, while facilitating interpretation by the researchers. Similarly, using multiple observers can produce reliability where objectivity is desired, instead of a more subjective interpretation. Scholta et al. (2020) create a model for decision support in personal project management (PPM) to address flexibility and context awareness. They engaged subjects in two types of evaluation consisting of experiments with two scenarios at the same time as multiple sources of evidence. They also included a questionnaire with open-ended questions to assess usability.

Reliability can also be shown by consistency in the results of multi-level observations; for example, observations of the performance of an artefact at both the organisational and individual levels. An example that demonstrates synchronic reliability of an artefact is the evaluation of the Cybergate prototype instantiation using an Enron email corpus (Abbasi & Chen, 2008).

Synchronic reliability can help establish that an artefact works dependably; its operation is robust. For example, evaluating a single instance of a prototype in an ideal setting could confirm validity, but demonstrate only weak reliability. If, instead, a single instance evaluation takes place in a setting at the extreme edge of the operational conditions (*contra-ideal* or worst expected condition) for the prototype's design, it can confirm both the validity and the reliability of the prototype. Such an evaluation is equivalent to a single test of an engineering artefact in proximity to its most likely failure mode.

Studies often evaluate the functionality of an instantiated prototype in a single, difficult case study. For example, through a single case study based on a financial services website, Albert et al. (2004) provided a real-world example of the implementation of their general model for gathering, inferring, segmenting and tracking online and offline visits to websites that could be used for customer relationship management. This demonstrates the model in a naturalistic setting, providing the most robust test of its utility.

5.6 | Measures: Diachronic reliability

Diachronic reliability refers to the stability of observations over time, or at different times (Kirk & Miller, 1986). The observations may be concerned with the performance of the designed artefact. In engineering, reliability is based on an artefact performing its purpose over time and under the conditions encountered (Barlow, 1998). It aligns with a diachronic view of artefact reliability and can be accomplished through measures such as persistent observation, triangulation or transferability using different approaches such as focus groups, interviews and questionnaires.

This kind of reliability can be established by repeating the measures of the artefact's performance and utility in different settings, with different users, on different problems and at different times. In this way, the utility of the artefact is shown to be independent of the unique context for which it was originally designed. Miah et al. (2014) provide a diachronic assessment of reliability of a collective decision support system artefact in an agricultural setting. Adomavicius et al. (2008) report on two case studies of their proposed artefacts for technology landscape analysis. The temporality aspect of these cases demonstrates the diachronic reliability of the artefact.

Diachronic reliability of *measures* indicates that any measurement of the artefact is reliable when used in different settings or at different times. The reliability under different scenarios demonstrates the applicability of the results beyond the current study, yielding confidence in its generalizability. Paralleling the concept of reliability, dependability, refers to the stability of the findings over time.

To establish this kind of artefact reliability, a study can demonstrate the execution of multiple instances of measurements or observations at different times and/or in different settings with consistency in the results obtained. The measures and observations can themselves require a degree of abstraction, generality or interpretation needed to surmount the uniqueness of settings and reveal essential consistency. Techniques that can be used for such artefact reliability claims include model-based observations in comparative or longitudinal case studies. For example,

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McLaren et al. (2011) developed and evaluated the reliability (and utility) of their artefact (a multi-level strategic fit measurement model), using a multiple case study.

Dincelli and Chengalur-Smith (2020) created an artefact for a gamified approach to security education longitudinal experiment. They tested it with multiple measures (e.g., attitude, intention) and reported reliability for the measurement constructs which were adapted from a prior scale. Akello et al. (2023) propose a blockchain-based model for gun control and tracing based on a chain-of-custody. They show multiple strategies to demonstrate the value of the artefact and demonstrate utility through different types of focus groups. They also used different measures to demonstrate the quality and efficiency of the design. In response for a need for governments, generally to be able to standardise forms, Scholta et al. (2020) design a 'meta-model for a reference form modelling language in governments' (p. 536) by applying the model of Peffers et al. (2007). Their artefact (meta-model) is intended to be useful for defining reference form models. They instantiated their model and used 3 cycles of evaluation for assessment of efficacy (with focus groups), completeness (with case studies) and effectiveness.

Piccoli et al. (2020) develop a sociotechnical artefact for providing feedback for digital skill accomplishment at scale. The researchers specifically state that they strive for diachronic reliability by analysing the performance of the artefact with different users on different problems, while being aware of false positives and false negatives (p. 116). The results led the researchers to conclude that their artefact was better than humans. This shows that this type of designation of reliability influenced the way in which reliability was assessed because it was identified at the beginning as appropriate.

5.7 | Design knowledge: Synchronic reliability

Synchronic reliability considers the degree to which design knowledge is dependable for a specific context. Synchronic reliability is often inferred from the idiographic methods used to produce it. Since idiographic design knowledge focuses on providing insights, the reliability of such knowledge considers how the knowledge was distilled from the phenomenon (Baskerville et al., 2015). Lincoln and Guba (1985) recommend that such knowledge be evaluated, not just for the knowledge itself, but also for the methods of its production. The latter may include contextualization, dialogical reasoning and sensitivity to multiple interpretations (Klein & Myers, 1999). To an extent, synchronic reliability of design knowledge implies that the knowledge is ready-to-use elsewhere; it is generalizable.

Synchronic reliability is established when the design knowledge leads to the discovery of novel and significant scientific findings and/or novel and important artefacts. Because synchronic studies focus on a singular instance of an artefact, reliability is shown in the design process through iteration, such as in prototyping. With each iteration, the reliability of the design knowledge is justified by the consistency of the results (in terms of additional, incremental, and valuable discoveries) in comparison with the results of prior iterations. Kohler et al. (2011), for example, present a virtual co-creation system, called Ideation Quest, which was created, deployed, evaluated, and improved in two action research cycles. The cycles took place over 2 years. The evaluation involved open coding and design principles related to co-creation adapted or added.

5.8 | Design knowledge: Diachronic reliability

The diachronic view of reliability of design knowledge focuses on whether the designer's knowledge, as embedded in the artefact, is useful over time (e.g., the principles of a particular design theory prove appropriate for making valuable artefacts in different projects). The diachronic view of design knowledge is concerned with the degree to which design theories or other forms of design knowledge (design principles, design models or design methods), can be applied to develop artefacts for other settings.

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The reliability of such knowledge, thus, lies in the successful application of the design knowledge in producing new artefacts for different settings, different design problems and different times. Establishing reliability typically involves using the design knowledge iteratively across different settings and/or problems. In other words, reliable design knowledge evolves by iteration in artefact use. The reliability is evident by the presence among the outcomes of each of the various iterations of important discoveries in design and/or science. An example of diachronically reliable design knowledge is the concept of a theory nexus originating in Carroll and Kellogg's (1989, p. 8) design theorising for a 'training wheels interface'. This design concept was adapted later as the basis for a design theory nexus in multiple cases, including organisational change design and user engagement design (Pries-Heje & Baskerville, 2008).

6 | DISCUSSION

Design is a complex and creative activity that arises in different disciplines (Baskerville et al., 2011). It is also fundamental to research in information systems (March & Storey, 2008). However, design science research is different from other types of research because of the existence of an artefact. The artefact must be developed (requiring reliability of the development method) and evaluated (requiring reliability of the evaluation measures). The development of the artefact leads to additions to the design knowledge base (requiring reliability of the design knowledge).

This research has proposed a *Design Science Reliability Framework*, intended to help design science researchers systematically consider the reliability of their work outcomes (artefacts and other knowledge contributions). The framework has three dimensions: what (artefact, method, measures, design knowledge), when (moment), and how (techniques). The framework has been illustrated by representative design science research projects. The intent of our framework, as is the purpose of all frameworks, is to provide a general, and useful, way to understand a phenomenon (Leonard et al., 2016); here, reliability in design science research.

It is always important to identify the qualities of an artefact that would make an important contribution to design science research. Evaluation is crucial because it demonstrates the utility and quality of the artefacts and knowledge outcomes, as well as the rigour of the research process used to obtain them. Although the importance of evaluating different types of design science research outcomes (artefacts, design knowledge) is well-recognised, reliability is seldom discussed to confirm the rigour of the studies or their assessment. This led to the need to distinguish the evaluation of the rigour of the research process (method and measures) from the evaluation of the quality of the outcomes (artefact and design knowledge).

6.1 | Extension of prior research on evaluation

The *Design Science Reliability Framework* extends prior work on evaluation in design science research, including that of Peffers et al. (2012) and Venable et al. (2016). It is intended to improve reliance on face validity generalities; for example, 'any appropriate empirical evidence or logical proof' (Peffers et al., 2007, p. 56). Instead, the framework requires the evaluation activities to consider whether such evidence can be considered reliable, which helps support the assumptions that design science research results have scientific standing. In design science research, this evidence is often embodied in the functionality and utility of the artefact as an outcome. Showing that this evidence (or 'near-proof') of functionality and utility has the character of being scientifically reliable, leading to claims about validity, can bolster the acceptability of its design principles or theory, into the body of received knowledge in design science research.

The direct outcomes of the framework should be of practical use. The identification of the importance of *what* and *when*, as well as the eight techniques (*how*), should help design scientists classify their evidence of reliability. The examples provided illustrate successful techniques in each category, which design scientists could use analogously to gain insights into how to conduct and present their own efforts to establish reliability. A design scientist must first recognise the kind of reliability (artefact, measures, method or design knowledge) needed. It should then be feasible to map the characteristics of a specific outcome to the appropriate kind of reliability assessment.

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TABLE 5 Establishment of different kinds of reliability.

Kind	Definition	Description
Synchronic artefact	Single assessment of artefact in original environment	Show reliability by assessing the utility of the artefact in environment in which design science research project initially required
Diachronic artefact	Repeated assessments of the artefact's performance and utility independent of the original environment	Show reliability by assessing the utility of the artefact in a natural environment
Synchronic measures	The practical adoption of the artefact could have utility	Artefact utility reliable in at least one instance. Research is pragmatic (means led to the ends) in one instance
Diachronic measures	The practical adoption of the artefact will have utility	Artefact utility reliable in a class of instances. Research ultimately pragmatic (the means led to the ends) in each instance in the class
Synchronic method	This research process is convincing	Method reliable in at least one instance; scientific and rigorous
Diachronic method	This research process is transferable to other times and settings	Method shown to be reliable in class of instances. In each instance, method was scientific and rigorous
Synchronic design knowledge	Scientific knowledge produced	Design knowledge reliable in at least one instance
Diachronic design knowledge	Scientific knowledge produced	Design knowledge reliable in class of instances

The accumulation of validation is 'never-ending' (Hambleton, 1984). The intended contribution of the framework is to serve as a useful way for design scientists to identify and appreciate how to establish reliability. Categorising reliability recommendations by the various facets of design science research (artefacts, measures, methods and design knowledge), is an attempt to provide reasonable and suitable guidance to realise better reliability than possible in the past and, subsequently, gain confidence in the validity of the research. Table 5 summarises the establishment of different kinds of reliability based on the framework.

6.2 | Application of framework

To understand how to identify which of the techniques might be needed and applied, consider the a framework for evaluation in design science research (FEDS) framework (Venable et al., 2016), which identifies four kinds of evaluation strategies for design science research. Table 6 shows a mapping of the strategies to the kinds of reliability presented in this paper and indicates the value added by establishing reliability in this way. Obviously, it is not feasible, or necessary, to show all eight kinds of reliability for a given study.

Overall, the framework can be used to guide design science researchers to establish reliability. Application of the techniques should be helpful to researchers in formulating their studies so they can establish or improve reliability.

6.3 | Future research

It would be useful to examine fundamental scientific concepts such as ontologies, epistemologies, methodology and axiology to gain additional supporting justification for the choice of the assessment approaches. There are known

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TABLE 6 Reliability for design science research evaluation strategies.

Evaluation strategy	Kinds of reliability	Value
• Quick and simple	• Synchronic measures	• Shows utility of the artefact in multi-level environment. Common in many design science research studies
• Human risk and effectiveness	• Synchronic measures • Diachronic measures • Synchronic design knowledge • Diachronic design knowledge	• Expands depth of understanding of the performance required in multiple social settings
• Technical risk and efficacy	• Synchronic measures • Diachronic measures	• Shows artefact consistently performs as intended
• Purely technical artefact strategy	• Synchronic methods • Synchronic measures • Synchronic design knowledge	• Emphasises how design science method, artefact and design knowledge work in at least one instance. Diminished concern of multiple times

aspects of design science research where challenges of reliability may be extremely difficult to address, for example, aesthetics, especially with respect to the user interface (Baskerville et al., 2018). The creativity of a design scientist is difficult to replicate, even though there has been much research carried out to understand creativity in various fields of research.

Much current design science research tends to be idiographic, focusing on a single instance of an artefact. Establishing scientific reliability in idiographic studies is quite different than establishing it in more nomothetic studies. Future research is needed to consider how to attend rigour throughout the life cycle of idiographic forms of design science.

Future research is also needed to show reliability in nonmaterial artefacts. Nonmaterial artefacts include methods, models and constructs. The *methods* in this sense are not usually scientific methods, but rather practical methods intended for utilitarian purposes outside of scientific research (March & Smith, 1995). Other kinds of theoretical formulations might also serve as possible artefacts. It is likely that techniques, similar to those proposed here for reliability in scientific methods, will help establish reliability in nonmaterial artefacts, although further investigation is needed.

Our research derived a *Design Science Reliability Framework* based on a conceptual argument grounded in logic and previously published work. Had this paper used a design science research approach, the framework might be considered an artefact, and future researchers might use a design science research approach to reflexively apply the framework for its own reliability assessment. In this way, the reliability of the design knowledge in the framework could be established by using multiple evaluation methods (synchronically) during a relevant practical project, or by developing evidence for the useful application of the knowledge in other design science projects (diachronically). A less resource-intensive approach would be to apply the design knowledge, retrospectively, to previous design science research projects (diachronically).[10]

The *Design Science Reliability Framework* is intended to be a useful addition to the toolbox of design science researchers who are encouraged to use it selectively, according to their projects and interests. Application of the framework to many research projects reported in additional publication outlets (other journals and related conferences) could be helpful in creating a repository of examples for future researchers. The paper focuses on a top-down approach to conducting design science research (inventing an artefact to address a generalised type of problem) and not inverse approaches that generalise after several cases have been documented for example, strategies of Iivari

[10]We thank an anonymous reviewer for this suggested observation.

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(2015); (Rohde et al., 2017). Further, this research does not consider other, potential bigger effects; for example, researchers are not addressing problems that are relevant or the proposed artefacts are not actually effective, irrespective of reliability.

7 | CONCLUSION

Reliability is important for the evaluation of any research endeavour, yet has different characteristics in design science research and appears to be under represented and reported. This paper has proposed a *Design Science Reliability Framework*, in which reliability is addressed in terms of what is reliable (artefacts, method, measures and design knowledge) and when reliability is assessed (synchronic versus diachronic evaluation). From the framework, eight techniques emerge for how to assess and demonstrate reliability for the artefact and knowledge products. Outside of the information systems discipline, the framework might be useful to computing and engineering design disciplines that focus specifically on artefacts.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest.

DATA AVAILABILITY STATEMENT

This is no additional data associated with this paper.

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REFERENCES

Abbasi, A., Albrecht, C., Vance, A., & Hansen, J. (2012). Metafraud: A meta-learning framework for detecting financial fraud. *MIS Quarterly*, *36*, 1293–1327.

Abbasi, A., & Chen, H. (2008). CyberGate: A design framework and system for text analysis of computer-mediated communication. *MIS Quarterly*, *32*(4), 811–837. https://doi.org/10.2307/25148873

Abbasi, A., Zhou, Y., Deng, S., & Zhang, P. (2018). Text analytics to support sense-making in social media: A language-action perspective. *MIS Quarterly*, *42*(2), 427–464.

Adipat, B., Zhang, D., & Zhou, L. (2011). The effects of tree-view based presentation adaptation on mobile web browsing. *MIS Quarterly*, *35*, 99–121.

Adomavicius, G., Bockstedt, J. C., Gupta, A., & Kauffman, R. J. (2008). Making sense of technology trends in the information technology landscape: A design science approach. *MIS Quarterly*, *32*(4), 779–809. https://doi.org/10.2307/25148872

Akello, P., Vemprala, N., Lang Beebe, N., & Raymond Choo, K.-K. (2023). Blockchain use case in ballistics and crime gun tracing and intelligence: Toward overcoming gun violence. *ACM Transactions on Management Information Systems*, *14*(1), 1–26.

Akoka, J., Comyn-Wattiau, I., Prat, N., & Storey, V. C. (2022). Knowledge contributions in design science research: Paths of knowledge types. *Decision Support Systems*, *166*, 113898.

Albert, T. C., Goes, P. B., & Gupta, A. (2004). GIST: A model for design and management of content and interactivity of customer-centric web sites. *MIS Quarterly*, *28*, 161–182.

13652575, 2025, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/isj.12564 by Southwestern University, Wiley Online Library on [18/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Anderson, C., Carvalho, A., Kaul, M., & Merhout, J. W. (2023). Blockchain innovation for consent self-management in health information exchanges. *Decision Support Systems*, *174*, 114021.

Arazy, O., Kumar, N., & Shapira, B. (2010). A theory-driven design framework for social recommender systems. *Journal of the Association for Information Systems*, *11*(9), 2–490.

Astor, P. J., Adam, M. T. P., Jercic, P., Schaaff, K., & Weinhardt, C. (2013). Integrating biosignals into information systems: A NeuroIS tool for improving emotion regulation. *Journal of Management Information Systems*, *30*(3), 247–277. https://doi.org/10.2753/mis0742-1222300309

Avdiji, H., & Winter, R. (2019). Knowledge gaps in design science research.

Barley, S. R. (1990). Images of imaging: Notes on doing longitudinal field work. *Organization Science*, *1*(3), 220–247.

Barlow, R. E. (1998). *Engineering reliability*. American Statistical Association and the Society for Industrial and Applied Mathematics.

Baskerville, R. (1996). Deferring generalizability four classes of generalization in social enquiry. *Scandinavian Journal of Information Systems*, *8*, 5–28.

Baskerville, R., Kaul, M., & Storey, V. C. (2011). Unpacking the duality of design science. *International Conference on Information Systems*. Shanghai, China, 4-7 December 2011, 15.

Baskerville, R., Kaul, M., & Storey, V. C. (2015). Genres of inquiry in design-science research: Justification and evaluation of knowledge production. *MIS Quarterly*, *39*(3), 541–564.

Baskerville, R., Kaul, M., & Storey, V. C. (2018). Aesthetics in design science research. *European Journal of Information Systems*, *1–14*, 140–153. https://doi.org/10.1080/0960085X.2017.1395545

Baskerville, R., & Myers, M. D. (2015). Design ethnography in information systems. *Information Systems Journal*, *25*(1), 23–46.

Baskerville, R., & Pries-Heje, J. (2014). Design theory Projectability. In B. Doolin, E. Lamprou, & N. Mitev (Eds.), *Information Systems and Global Assemblages: (Re)Configuring Actors, Artefacts, Organizations: Proceedings of the IFIP Working Group 8.2 Working Conference* (pp. 219–232). Springer.

Baskerville, R., Vom Brocke, J., Mathiassen, L., & Scheepers, H. (2023). Clinical research from information systems practice. *European Journal of Information Systems*, *32*(1), 1–9.

Baskerville, R. L., Kaul, M., & Storey, V. C. (2017). *Establishing reliability in design science research*. ICIS.

Baskerville, R. L., & Pries-Heje, J. (2019). Projectability in design science research. *J Inf Technol Theory Appl*, *20*(1), 3.

Bateson, M., & Martin, P. (2021). *Measuring behaviour: An introductory guide*. Cambridge university press.

Bera, P., Burton-Jones, A., & Wand, Y. (2011). Guidelines for designing visual ontologies to support knowledge identification. *MIS Quarterly*, *35*, 883–908.

Boudreau, M.-C., Gefen, D., & Straub, D. W. (2001). Validation in information systems research: A state-of-the-art assessment. *MIS Quarterly*, *25*, 1–16.

Boughzala, I., & De Vreede, G.-J. (2015). Evaluating team collaboration quality: The development and field application of a collaboration maturity model. *Journal of Management Information Systems*, *32*(3), 129–157.

Brandt, T., Feuerriegel, S., & Neumann, D. (2018). Modeling interferences in information systems design for cyberphysical systems: Insights from a smart grid application. *European Journal of Information Systems*, *27*(2), 207–220.

Brendel, A. B., Lembcke, T.-B., Muntermann, J., & Kolbe, L. M. (2021). Toward replication study types for design science research. *Journal of Information Technology*, *36*(3), 198–215.

Brendel, A. B., & Muntermann, J. (2022). Replication of design theories: Reflections on function, outcome, and impact. *Information Systems Journal*, *32*, 1158–1176.

Briggs, R. O., Kolfschoten, G. L., de Vreede, G.-J., Lukosch, S., & Albrecht, C. C. (2013). Facilitator-in-a-box: Process support applications to help practitioners realize the potential of collaboration technology. *Journal of Management Information Systems*, *29*(4), 159–194.

Burrell, G., & Morgan, G. (1979). *Sociological paradigms and organisational analysis* (Vol. 248). Heinemann.

Burton-Jones, A., Wand, Y., & Weber, R. (2009). Guidelines for empirical evaluations of conceptual modeling grammars. *Journal of the Association for Information Systems*, *10*(6), 1–532.

Butler, B. S., & Gray, P. H. (2006). Reliability, mindfulness, and information systems. *MIS Quarterly*, *30*, 211–224.

Bygstad, B. (2017). Generative innovation: A comparison of lightweight and heavyweight IT. *Journal of Information Technology*, *32*(2), 180–193.

Carmines, E., & Zeller, R. (1979). *Reliability and validity assessment* (Vol. 07-017). Sage Publications.

Carroll, J., & Kellogg, W. (1989). Artifact as theory-nexus: Hermeneutics meets theory-based design. Proceedings of the SIGCHI conference on Human factors in computing systems.

Carvalho, J. Á. (2012). *Validation criteria for the outcomes of design research IT*. Artefact Design & Workpractice Intervention.

Chanson, M., Bogner, A., Bilgeri, D., Fleisch, E., & Wortmann, F. (2019). Blockchain for the IoT: Privacy-preserving protection of sensor data. *Journal of the Association for Information Systems*, *20*(9), 1274–1309.

13652575, 2025, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/isj.12564 by Southwestern University, Wiley Online Library on [18/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chen, R., Sharman, R., Rao, H. R., & Upadhyaya, S. J. (2013). Data model development for fire related extreme events: An activity theory approach. *MIS Quarterly*, *37*(1), 125–147. http://search.ebscohost.com/login.aspx?direct=true&AuthType=ip,url&db=bth&AN=85634428&site=ehost-live

Coenen, T., Coertjens, L., Vlerick, P., Lesterhuis, M., Mortier, A. V., Donche, V., Ballon, P., & De Maeyer, S. (2018). An information system design theory for the comparative judgement of competences. *European Journal of Information Systems*, *27*(2), 248–261.

Cook, T. D., Campbell, D. T., & Day, A. (1979). *Quasi-experimentation: Design & analysis issues for field settings* (Vol. 351). Rand McNaly College Publishing.

Cross, N. (1982). Designerly Ways of Knowing. *Design Studies*, *3*(4), 221–227. https://doi.org/10.1016/0142-694X(82)90040-0

Dadgar, M., & Joshi, K. D. (2018). The role of information and communication technology in self-management of chronic diseases: An empirical investigation through value sensitive design. *Journal of the Association for Information Systems*, *19*(2), 2–112.

Day, J. M., Junglas, I., & Silva, L. (2009). Information flow impediments in disaster relief supply chains. *Journal of the Association for Information Systems*, *10*(8), 1–660.

De Leoz, G., & Petter, S. (2018). Considering the social impacts of artefacts in information systems design science research. *European Journal of Information Systems*, *27*(2), 154–170.

De Schryver, M., Hughes, S., Rosseel, Y., & De Houwer, J. (2016). Unreliable yet still replicable: A comment on LeBel and Paunonen (2011). *Frontiers in Psychology*, *6*, 173501.

Delbridge, R., & Fiss, P. C. (2013). *Editors' comments: Styles of theorizing and the social organization of knowledge*. In (Vol. 38, pp. 325–331). Academy of Management Briarcliff.

Deng, Q., Wang, Y., & Ji, S. (2017). Design science research in information systems: A systematic literature review 2001–2015. In *International conference on information resources management (CONF-IRM)*. Association for Information Systems (AIS).

Dieter, G. E., & Schmidt, L. C. (2009). *Engineering Design*. McGraw-Hill Higher Education.

Dincelli, E., & Chengalur-Smith, I. (2020). Choose your own training adventure: Designing a gamified SETA artefact for improving information security and privacy through interactive storytelling. *European Journal of Information Systems*, *29*(6), 669–687.

Doty, D. H., & Glick, W. H. (1994). Typologies as a unique form of theory building: Toward improved understanding and modeling. *Academy of Management Review*, *19*(2), 230–251.

Ebel, P., Bretschneider, U., & Leimeister, J. M. (2016). Leveraging virtual business model innovation: A framework for designing business model development tools. *Information Systems Journal*, *26*(5), 519–550.

Fahmideh, M., Daneshgar, F., Rabhi, F., & Beydoun, G. (2019). A generic cloud migration process model. *European Journal of Information Systems*, *28*(3), 233–255.

Garner, J. P. (2005). Stereotypies and other abnormal repetitive behaviors: Potential impact on validity, reliability, and replicability of scientific outcomes. *ILAR Journal*, *46*(2), 106–117.

Germonprez, M., Hovorka, D., & Collopy, F. (2007). A theory of tailorable technology design. *Journal of the Association for Information Systems*, *8*(6), 21–367.

Gomber, P., Kauffman, R. J., Parker, C., & Weber, B. W. (2018). On the fintech revolution: Interpreting the forces of innovation, disruption, and transformation in financial services. *Journal of Management Information Systems*, *35*(1), 220–265.

Gosain, S., Malhotra, A., & El Sawy, O. A. (2004). Coordinating for flexibility in e-business supply chains. *Journal of Management Information Systems*, *21*(3), 7–45.

Gottschalk, S., Parvez, S., Yigitbas, E., & Engels, G. (2022). Designing platforms for crowd-based software prototype validation: A design science study. In *International Conference on Product-Focused Software Process Improvement* (pp. 334–350). Springer International Publishing.

Gregor, S., Chandra Kruse, L., & Seidel, S. (2020). Research perspectives: The anatomy of a design principle. *Journal of the Association for Information Systems*, *21*(6), 2–1652.

Guba, E. (1981). Criteria for assessing the trustworthiness of naturalistic inquiries. *Educational Technology Research and Development*, *29*(2), 75–91. https://doi.org/10.1007/bf02766777

Guba, E. G., & Lincoln, Y. S. (1994). Competing paradigms in qualitative research. *Handbook of Qualitative Research*, *2*(163–194), 105.

Gummesson, E. (1991). *Qualitative methods in management research*. Sage Publications.

Gummesson, E. (2000). *Qualitative methods in management research*. Sage.

Hambleton, R. K. (1984). Validating the test scores. In R. A. Beck (Ed.), *A guide to criterion-referenced test construction* (pp. 199–230). *John Hopkins University Press*.

Holmström, J., Ketokivi, M., & Hameri, A. P. (2009). Bridging practice and theory: A design science approach. *Decision Sciences*, *40*(1), 65–87.

Hooker, J. N. (2004). Is design theory possible? *Journal of Information Technology Theory and Application (JITTA)*, *6*(2), 8.

Iivari, J. (2015). Distinguishing and contrasting two strategies for design science research. *European Journal of Information Systems*, *24*(1), 107–115.

13652575, 2025, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/isj.12564 by Southwestern University, Wiley Online Library on [18/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Iivari, J., Rotvit Perlt Hansen, M., & Haj-Bolouri, A. (2021). A proposal for minimum reusability evaluation of design principles. *European Journal of Information Systems*, *30*(3), 286–303.
Jaakkola, E. (2020). Designing conceptual articles: Four approaches. *AMS Review*, *10*(1–2), 18–26.
Johannesson, P., & Perjons, E. (2014). *An introduction to design science* (Vol. 10). Springer.
John, B. M., Chua, A. Y., Goh, D. H. L., & Wickramasinghe, N. (2016). Graph-based cluster analysis to identify similar questions: A design science approach. *Journal of the Association for Information Systems*, *17*(9), 2.
Kartseva, V., Hulstijn, J., Gordijn, J., & Tan, Y.-H. (2010). Control patterns in a health-care network. *European Journal of Information Systems*, *19*, 320–343.
Keith, M., Demirkan, H., & Goul, M. (2013). Service-oriented methodology for systems development [article]. *Journal of Management Information Systems*, *30*(1), 227–259. <Go to ISI>://WOS:000323353200008.
Kellert, S. H., Longino, H. E., & Waters, C. K. (2006). *Scientific Pluralism* (Vol. 19). U of Minnesota Press.
Kerlinger, F. N. (1973). *Foundations of behavioral research*. Holt.
Kirk, J., & Miller, M. L. (1986). *Reliability and validity in qualitative research* (Vol. 1). Sage Publications.
Klein, H. K., & Myers, M. D. (1999). A set of principles for conducting and evaluating interpretive field studies in information systems. *MIS Quarterly*, *23*(1), 67–93.
Kohler, T., Fueller, J., Matzler, K., Stieger, D., & Füller, J. (2011). Co-creation in virtual worlds: The design of the user experience. *MIS Quarterly*, *35*(3), 773–788. https://doi.org/10.2307/23042808
Kuechler, W., & Vaishnavi, V. (2012). A framework for theory development in design science research: Multiple perspectives. *Journal of the Association for Information Systems*, *13*(6), 395–423.
Lacity, M., Willcocks, L., & Gozman, D. (2021). Influencing information systems practice: The action principles approach applied to robotic process and cognitive automation. *Journal of Information Technology*, *36*(3), 216–240.
Larsen, K. R., Lukyanenko, R., Mueller, R. M., Storey, V. C., VanderMeer, D., Parsons, J., & Hovorka, D. S. (2020). Validity in design science research. In *Designing for digital transformation. Co-creating services with citizens and industry: 15th international conference on design science research in information systems and technology, DESRIST 2020, Kristiansand, Norway, December 2–4, 2020, Proceedings 15* (pp. 272–282). Springer International Publishing.
Lee, J. S., Pries-Heje, J., & Baskerville, R. (2011). Theorizing in design science research. In *Service-oriented perspectives in design science research: 6th international conference, DESRIST 2011, Milwaukee, WI, USA, May 5-6, 2011, Proceedings 6* (pp. 1–16). Springer Berlin Heidelberg.
Leonard, A., Miller, S. A., Christopher, R., Gleason, K., & Franklin, R. (2016). Theories in the field of community psychology. *Global Journal of Community Psychology Practice*, *7*(2), 1–9.
Lincoln, Y. S., & Guba, E. G. (1985). *Naturalistic inquiry* (Vol. 75). SAGE Publications, Incorporated.
Liu, D., Santhanam, R., & Webster, J. (2017). Toward meaningful engagement. *MIS Quarterly*, *41*(4), 1011–1034.
Liu, X., Wang, G. A., Fan, W., & Zhang, Z. (2020). Finding useful solutions in online knowledge communities: A theory-driven design and multilevel analysis. *Information Systems Research*, *31*(3), 731–752.
Lukyanenko, R., Evermann, J., & Parsons, J. (2015). Guidelines for establishing instantiation validity in IT artifacts: A survey of IS research. In *New Horizons in Design Science: Broadening the Research Agenda: 10th International Conference, DESRIST 2015, Dublin, Ireland, May 20-22, 2015, Proceedings 10*. Springer-Verlag.
Lukyanenko, R., Maass, W., & Storey, V. C. (2022). Trust in artificial intelligence: From a foundational trust framework to emerging research opportunities. *Electronic Markets*, *32*(4), 1993–2020.
March, S. T., & Smith, G. F. (1995). Design and natural science research on information technology. *Decision Support Systems*, *15*(4), 251–266. https://doi.org/10.1016/0167-9236(94)00041-2
March, S. T., & Storey, V. C. (2008). Design science in the information systems discipline: An introduction to the special issue on design science research. *MIS Quarterly*, *32*, 725–730.
Markus, M. L., Majchrzak, A., & Gasser, L. (2002). A design theory for systems that support emergent knowledge processes. *MIS Quarterly*, *26*(3), 179–212.
Matook, S., & Brown, S. A. (2017). Characteristics of IT artifacts: A systems thinking-based framework for delineating and theorizing IT artifacts. *Information Systems Journal*, *27*(3), 309–346.
McLaren, T. S., Head, M. M., Yuan, Y., & Chan, Y. E. (2011). A multilevel model for measuring fit between a firm's competitive strategies and information systems capabilities. *MIS Quarterly*, *35*(4), 909–929.
Meth, H., Mueller, B., & Maedche, A. (2015). Designing a requirement mining system. *Journal of the Association for Information Systems*, *16*(9), 2–837.
Mettler, T., & Winter, R. (2016). Are business users social? A design experiment exploring information sharing in enterprise social systems. *Journal of Information Technology*, *31*(2), 101–114.
Miah, S. J., Kerr, D., & Hellens, L. v. (2014). A collective artefact design of decision support systems: Design science research perspective. *Information Technology & People*, *27*(3), 259–279. https://doi.org/10.1108/ITP-04-2012-0041
Morana, S., Kroenung, J., Maedche, A., & Schacht, S. (2019). Designing process guidance systems. *Journal of the Association for Information Systems*, *20*(5), 6–535.

13652575, 2025, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/isj.12564 by Southwestern University, Wiley Online Library on [18/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Müller, S. D., Mathiassen, L., & Saunders, C. (2021). Pluralist theory building: A methodology for generalizing from data to theory*. In *Advancing Information Systems Theories* (pp. 309–361). Rationale and Processes.

Närman, P., Holm, H., Ekstedt, M., & Honeth, N. (2013). Using enterprise architecture analysis and interview data to estimate service response time. *The Journal of Strategic Information Systems*, *22*(1), 70–85.

Nunamaker, J. J. F., Briggs, R. O., Derrick, D. C., & Schwabe, G. (2015). The last research mile: Achieving both rigor and relevance in information systems research. *Journal of Management Information Systems*, *32*(3), 10–47.

Oberländer, A. M., Röglinger, M., Rosemann, M., & Kees, A. (2018). Conceptualizing business-to-thing interactions–A sociomaterial perspective on the internet of things. *European Journal of Information Systems*, *27*(4), 486–502.

Oruç, S., Eren, P. E., & Koçyiğit, A. (2022). A constraint programming model for making recommendations in personal process management: A design science research approach. *Decision Support Systems*, *152*, 113665.

Oyekunle, D. O. T., Okwudili Matthew, U., Preston, D., & Boohene, D. (2024). Trust beyond technology algorithms: A theoretical exploration of consumer trust and behavior in technological consumption and AI projects. *Journal of Computer and Communications*, *12*(6), 72–102.

Pant, G., & Srinivasan, P. (2013). Status locality on the web: Implications for building focused collections. *Information Systems Research*, *24*(3), 802–821.

Parsons, J., & Wand, Y. (2008). Using cognitive principles to guide classification in information systems modeling. *MIS Quarterly*, *32*, 839–868.

Parsons, J., & Wand, Y. (2013). Extending classification principles from information modeling to other disciplines. *Journal of the Association for Information Systems*, *14*(5), 245–273.

Peffers, K., Rothenberger, M., Tuunanen, T., & Vaezi, R. (2012). Design science research evaluation. In K. Peffers, M. Rothenberger, & B. Kuechler (Eds.), *Advances in theory and practice: 7th international conference, DESRIST 2012, Las Vegas, NV, USA, May 14–15, 2012. Proceedings* (pp. 398–410). Springer. https://doi.org/10.1007/978-3-642-29863-9_29

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, *24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2008). A design science research methodology for information systems research. *Journal of Management Information Systems*, *24*(3), 45–77.

Pentland, B., & Feldman, M. (2008). Designing routines: On the folly of designing artifacts, while hoping for patterns of action. *Information and Organization*, *18*(4), 235–250.

Piccoli, G., Rodriguez, J., Palese, B., & Bartosiak, M. L. (2020). Feedback at scale: Designing for accurate and timely practical digital skills evaluation. *European Journal of Information Systems*, *29*(2), 114–133.

Pillet, J. C., Carillo, K. D., Vitari, C., & Pigni, F. (2023). Improving scale adaptation practices in information systems research: Development and validation of a cognitive validity assessment method. *Information Systems Journal*, *33*(4), 842–889.

Posey, C., Raja, U., Crossler, R. E., & Burns, A. (2017). Taking stock of organisations' protection of privacy: Categorising and assessing threats to personally identifiable information in the USA. *European Journal of Information Systems*, *26*(6), 585–604.

Prat, N., Comyn-Wattiau, I., & Akoka, J. (2015). A taxonomy of evaluation methods for information systems artifacts. *Journal of Management Information Systems*, *32*(3), 229–267.

Pries-Heje, J., & Baskerville, R. (2008). The design theory nexus. *MIS Quarterly*, *32*(4), 731–755.

Ptaszynski, M., Lempa, P., Masui, F., Kimura, Y., Rzepka, R., Araki, K., Wroczynski, M., & Leliwa, G. (2019). Brute-force sentence pattern extortion from harmful messages for cyberbullying detection. *Journal of the Association for Information Systems*, *20*(8), 1075–1127.

Ralph, P., & Wand, Y. (2009). A proposal for a formal definition of the design concept. In K. Lyytinen, et al. (Eds.), *Design requirements workshop, LNBIP 14* (pp. 103–136). Springer-Verlag.

Rogers, T. B. (1995). *The psychological testing enterprise: An introduction*. Thomson Brooks/Cole Publishing Co.

Rohde, M., Brödner, P., Stevens, G., Betz, M., & Wulf, V. (2017). Grounded Design-A praxeological IS research perspective. *Journal of Information Technology*, *32*(2), 163–179.

Rosenkranz, C., Holten, R., Räkers, M., & Behrmann, W. (2017). Supporting the design of data integration requirements during the development of data warehouses: A communication theory-based approach. *European Journal of Information Systems*, *26*, 84–115.

Rossi, M., Ramesh, B., Lyytinen, K., & Tolvanen, J.-P. (2004). Managing evolutionary method engineering by method rationale. *Journal of the Association for Information Systems*, *5*(9), 12–391.

Runyan, W. (1983). Idiographic goals and methods in the study of lives. *Journal of Personality*, *51*(3), 413–437.

Sahoo, N., Singh, P. V., & Mukhopadhyay, T. (2012). A hidden Markov model for collaborative filtering. *MIS Quarterly*, *36*, 1329–1356.

13652575, 2025, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/isj.12564 by Southwestern University, Wiley Online Library on [18/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Schmitz, K., & Storey, V. C. (2020). Empirical test guidelines for content validity: Wash, rinse, and repeat until clean. *Communications of the Association for Information Systems*, *47*(1), 64–850.

Scholta, H., Balta, D., Räckers, M., Becker, J., & Krcmar, H. (2020). Standardization of forms in governments: A meta-model for a reference form modeling language. *Business & Information Systems Engineering*, *62*, 535–560.

Sein, M. K., Henfridsson, O., Purao, S., Rossi, M., Lindgren, R., & University of, B., School of, B., & It. (2011). Action design research. *Management Information Systems*, *35*(1), 37–56.

Shapira, Z. (2011). "I've got a theory paper—Do you?": Conceptual, empirical, and theoretical contributions to knowledge in the organizational sciences. *Organization Science*, *22*(5), 1312–1321.

Silic, M., & Lowry, P. B. (2020). Using design-science based gamification to improve organizational security training and compliance. *Journal of Management Information Systems*, *37*(1), 129–161.

Simon, H. A. (1969). *The Sciences of the Artificial*. MIT Press.

Sonnenberg, C., & Vom Brocke, J. (2012). Evaluations in the science of the artificial–reconsidering the build-evaluate pattern in design science research. Design science research in information systems. Advances in Theory and Practice: 7th International Conference, DESRIST 2012, Proceedings 7, Las Vegas, NV, USA.

Straub, D., Boudreau, M.-C., & Gefen, D. (2004). Validation guidelines for IS positivist research. *Communications of the Association for Information Systems*, *13*(1), 24.

Swinburne, R. (2001). *Epistemic justification*. Clarendon Press.

Tan, C.-W., Benbasat, I., & Cenfetelli, R. T. (2013). IT-mediated customer service content and delivery in electronic governments: An empirical investigation of the antecedents of service quality. *MIS Quarterly*, *37*, 77–109.

Thomas, M. A., Li, Y., & Lee, A. S. (2022). Generalizing the information systems artifact. *Information Systems Research*, *33*(4), 1452–1466.

Thoring, K., Mueller, R. M., Lecuna, A., & Badke-Schaub, P. (2023). *A framework of design method corroboration* 56th Hawaii international conference on system sciences.

Trochim, W. (2006). Qualitative Validity. Web Center For Social Research Methods. *Web Center for Social Research Methods*. (http://www.socialresearchmethods.net/kb/introval.php)

Twyman, N. W., Lowry, P. B., Burgoon, J. K., & Nunamaker, J. F., Jr. (2014). Autonomous scientifically controlled screening systems for detecting information purposely concealed by individuals. *Journal of Management Information Systems*, *31*(3), 106–137.

Vaishnavi, V. K., & Kuechler, W. (2015). *Design science research methods and patterns: Innovating information and communication technology*. CRC Press.

Venable, J., & Baskerville, R. (2012). Eating our own cooking: Toward a more rigorous design science of research methods. *Electronic Journal of Business Research Methods*, *10*(2), 141–153.

Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: A framework for evaluation in design science research. *European Journal of Information Systems*, *25*(1), 77–89. https://doi.org/10.1057/ejis.2014.36

Venkatesh, V., Aloysius, J. A., Hoehle, H., & Burton, S. (2017). Design and evaluation of auto-ID enabled shopping assistance artifacts in customers' mobile phones. *MIS Quarterly*, *41*(1), 83–114.

Vincenti, W. (1990). *What engineers know and how they know it–analytical studies from aeronautical history*. In: Johns Hopkins University Press.

Vom Brocke, J., Winter, R., Hevner, A., & Maedche, A. (2020). Special issue editorial–accumulation and evolution of design knowledge in design science research: A journey through time and space. *Journal of the Association for Information Systems*, *21*(3), 9–544.

Walls, J. G., Widmeyer, G. R., & El Sawy, O. A. (1992). Building an information system design theory for vigilant EIS. *Information Systems Research*, *3*(1), 36–59.

Wang, J., Ipeirotis, P. G., & Provost, F. (2017). Cost-effective quality assurance in crowd labeling. *Information Systems Research*, *28*(1), 137–158.

Wang, Q., Li, B., & Singh, P. V. (2018). Copycats vs. original mobile apps: A machine learning copycat-detection method and empirical analysis. *Information Systems Research*, *29*(2), 273–291.

Williams, K., Chatterjee, S., & Rossi, M. (2008). Design of emerging digital services: A taxonomy. *European Journal of Information Systems*, *17*(5), 505–517.

Woo, C., Saghafi, A., & Rosales, A. (2014). *What is a contribution to IS design science knowledge?* International Conference on Information Systems.

Xu, J., Wang, G. A., Li, J., & Chau, M. (2007). Complex problem solving: Identity matching based on social contextual information. *Journal of the Association for Information Systems*, *8*(10), 31–545.

Yang, Y., Singhal, S., & Xu, Y. (2012). Alternate strategies for a win-win seeking agent in agent-human negotiations. *Journal of Management Information Systems*, *29*(3), 223–256.

Yin, R. K. (1994). Discovering the future of the case study. Method in Evaluation Research. *Evaluation Practice*, *15*(3), 283–290.

13652575, 2025, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/isj.12564 by Southwestern University, Wiley Online Library on [18/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Zhou, S., Qiao, Z., Du, Q., Wang, G. A., Fan, W., & Yan, X. (2018). Measuring customer agility from online reviews using big data text analytics. *Journal of Management Information Systems*, *35*(2), 510–539.

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APPENDIX A

TABLE A1 Representative papers that mention reliability.

Paper	Reliability
Kartseva et al. (2010)	Uses or demonstrates synchronic reliability
Arazy et al. (2010)	Reliability with respect to empirical analysis for evaluation
Yang et al. (2012)	Tested with respect to validity for subjective dependent variables
Mettler and Winter (2016)	Focused on design; measurement validity and reliability
Twyman et al. (2014)	Examines autonomous scientifically controlled screening systems (ASCSS); build a prototype to evaluate and enhance design principles; mention reliability with respect to prediction accuracy
Ptaszynski et al. (2019)	With respect to pattern-based classifiers, for the version used, did not reach statistical significance decreasing reliability of n-gram based scores
Bygstad (2017)	Lightweight information technology (smart phones, tablets) versus heavyweight technology with latter requiring reliability
Chanson et al. (2019)	Veracity has varying degrees of reliability for sensor data sources; prototype evaluations include reliability
Sahoo et al. (2012)	Develop model, reliability with respect to learned transition probably matrix
Matook and Brown (2017)	Framework for articulating and theorising the IT artefact; create conceptual framework in the form of an ontology
Coenen et al. (2018)	Concern over use of reliability in assessment of current educational instruments; comparative judgement kernel theory; developed algorithm and tested reliability of system as a whole; information system design theory; argued to evaluate competencies
Xu et al. (2007)	Reliability mentioned with respect to different kinds of personal information
Oberländer et al. (2018)	Developed taxonomy for B2T (business to thing) patterns and assessed reliability and validity via Q-sort
Kartseva et al. (2010)	Usefulness and adequacy of control patterns in health case; need for reliability recognised in online feedback
Wang et al. (2018)	Design of apps; machine learning application
Wang et al. (2017)	Reliability with respect to crowd sourcing systems
Abbasi and Chen (2008)	Cybergate prototype artefact instantiation; inter-coding reliability reported for topic categorisation
Dadgar and Joshi (2018)	Information technology for self-management of diabetes as chronic disease; design of such systems; reliability with respect to trust
Chen et al. (2013)	Identified metadata as important for ensuring reliability, derived data elements as reliability numeric (content accuracy)

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TABLE A1 (Continued)

Paper	Reliability
Day et al. (2009)	Study of information flows in disaster relief supply chains; design principles; unreliability identified as information flow impediment
Venkatesh et al. (2017)	Design and compare shopping assistant artefacts; reliability and validity of measurement scales
Williams et al. (2008)	Reliability and value mentioned in four stages of digital service maturity (Table 3, p. 510)
Oruç et al. (2022)	Proof of concept, use, and value
Ebel et al. (2016)	Report inter-rater reliability for business model case studies
Boughzala and De Vreede (2015)	Measure quality of collaboration across boundaries; present collaboration maturity model; reliability with respect to product reliability
Briggs et al. (2013)	Create collaboration support system to reduce development cycles, enable nonprogrammers to design and develop systems, and package expertise; Cronbach's alpha reported on tool difficulty measure
Piccoli et al. (2020)	Sociotechnical artefact to provide feedback on digital skills mastery; strives for scalability, validity and reliability; specifically address reliability using diachronic reliability approach
Liu et al. (2020)	Developed text analytic framework and tested on companies; addressed reliability of manual annotation process in evaluation; mention cognitive processes related to content and reliability of messages in general sense
Gosain et al. (2004)	Propose design principles; reliability of model constructs reported
Bera et al. (2011)	Focus on design guidelines for visual ontologies; reliability for measures
Burton-Jones et al. (2009)	Assessment of conceptual model classification; heuristics for reliability with respect to coding
Astor et al. (2013)	Measurement scale applied has high reliability during evaluation
Keith et al. (2013)	Project success measured subjectively, one variable measured and reliability reported (Cronback's alpha); limited measures
Rossi et al. (2004)	Distinguish method from design rationale; state method changes in development environments necessary to improve 'performance, predictability, and reliability of methods' (p. 362)
Markus et al. (2002)	Reliability of knowledge base
Zhou et al. (2018)	Proposes alternative assessment to survey to avoid reliability problems (e.g., unsystematic and inadequate sampling, low respondent rates, bias), p. 534
Abbasi et al. (2012)	Performed *t*-test; claim consistent with prior work on generalisation and reliability
Meth et al. (2015)	Propose design theory; inter-relater reliability assessed manually and reported
Brandt et al. (2018)	Reliability mentioned with respect to EV car competing in energy market
Morana et al. (2019)	Reliability tested in measurement model
Gomber et al. (2018)	Design in finance industry; fintech; reliability only mentioned with respect to a start up; not true design science article
Parsons and Wand (2013)	Note that reliability of classes requires stability of inferences, with duration for the stability depending upon the context (nature of the phenomena) and classification purpose (p. 250). Systems analysis and design research
Müller et al. (2021)	Reliability for intercoder ratings
Rosenkranz et al. (2017)	Design to support data integration; reliability with respect to case study
Silic and Lowry (2020)	Created gamification security strategy; good reliability for measurement model reported and convergent and discriminant validity

(Continues)

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TABLE A1 (Continued)

Paper	Reliability
Vom Brocke et al. (2020)	Category of technology requirements
Pant and Srinivasan (2013)	Design of crawlers that focus on topical and status locality properties; reliability with respect to features of crawlers
Rosenkranz et al. (2017)	Communication theory for design theory development; reliability with respect to case studies
Posey et al. (2017)	Create a taxonomy for breaches; not presented as artefact in design science sense; ANOVA reported
Tan et al. (2013)	Classify services for government; theory of e-government service quality not presented as design science; construct validity reported
Abbasi et al. (2018)	Framework for sense making of online text; reported Cronbach's alphas
Adipat et al. (2011)	Descriptive statistics of perceived ease of use and usefulness presented via Cronbach's alpha (p. 112)
Närman et al. (2013)	Mention reliability with respect to measurement tool and threats to validity
John et al. (2016)	Propose graph-based cluster analysis method; combinations of cluster analysis to establish validity and reliability; checked subclusters
Yang et al. (2012)	Negotiation agent with two strategies (simultaneous-equivalent offers and delayed acceptance); address reliability with respect to their experimental design
McLaren et al. (2011)	Design MSF measurement model and evaluate its reliability using multiple case studies

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